# Xiuyuan Ding

Postdoctoral Scholar
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## RESEARCH INTERESTS

Atmospheric dynamics, stratosphere-troposphere coupling, weather and climate extremes, subseasonal-to-seasonal (S2S) predictability, climate variability and change, AI for weather and climate.

## EDUCATION

**University of California, Los Angeles** Sep 2019–Jun 2024
Ph.D. in Atmospheric and Oceanic Sciences, June 2024
M.S. in Atmospheric and Oceanic Sciences, June 2021
Advisor: Prof. Gang Chen

**Lanzhou University** Sep 2015–Jun 2019
B.S. in Atmospheric Sciences, June 2019

## ACADEMIC APPOINTMENTS

**Postdoctoral Scholar** Feb 2026–present
*University of California, Los Angeles*
AI-based weather prediction, stratospheric extreme events, surface temperature extremes.

**Postdoctoral Research Associate** Sep 2024–Feb 2026
*Princeton University and NOAA Geophysical Fluid Dynamics Laboratory*
Stratospheric influences on S2S prediction, subseasonal fire-weather prediction.

**Summer Intern** Jun–Sep 2024
*Lawrence Livermore National Laboratory*
Wildfire plumes in a regionally refined storm-resolving model.

## PUBLICATIONS

### Peer-reviewed

8. **Ding, X.**, Xiang, B., Chen, G., & Wang, L. (2026). Stratospheric wave predictability enhances surface forecasts over North America. *Science Advances*, 12(38), eaee4171. doi:10.1126/sciadv.aee4171
7. Xiang, B., Harris, L., Zhang, G., & **Ding, X.** (2026). Dual physical pathways shape subseasonal-to-seasonal predictability of North American summer heat extremes. *Communications Earth & Environment*, Advance online publication. doi:10.1038/s43247-026-03991-1
6. Chen, G., & **Ding, X.** (2026). Benchmarking the dynamic modes of stratosphere–troposphere coupling in Northern Annular Modes (NAM) in CMIP6 models. *Geophysical Research Letters*, 53(4), e2025GL121278. doi:10.1029/2025GL121278

5. **Ding, X.**, Chen, G., Wang, Y., & Sun, L. (2025). Demystifying the drivers of the spring warming asymmetry between Eurasia and North America. *Science Advances*, 11(22), eadu2364. doi:10.1126/sciadv.adu2364
4. **Ding, X.**, Chen, G., & Magnusdottir, G. (2024). North American cooling signature of strong stratospheric wave events depends on the QBO phase. *Environmental Research: Climate*, 3(3), 031006. doi:10.1088/2752-5295/ad53f6
3. **Ding, X.**, Chen, G., & Ma, W. (2023). Stratosphere–troposphere coupling of extreme stratospheric wave activity in CMIP6 models. *Journal of Geophysical Research: Atmospheres*, 128(16), e2023JD038811. doi:10.1029/2023JD038811 ***[Editor's Highlight]***
2. **Ding, X.**, Chen, G., Zhang, P., Domeisen, D. I. V., & Orbe, C. (2023). Extreme stratospheric wave activity as harbingers of cold events over North America. *Communications Earth & Environment*, 4, 187. doi:10.1038/s43247-023-00845-y
1. **Ding, X.**, Chen, G., Sun, L., & Zhang, P. (2022). Distinct North American cooling signatures following the zonally symmetric and asymmetric modes of winter stratospheric variability. *Geophysical Research Letters*, 49(6), e2021GL096076. doi:10.1029/2021GL096076

### In Prep/Under Review

3. **Ding, X.**, Chen, G., Xiang, B., & Nie, Y. (2026). Correcting the stratosphere improves S2S forecasts in NeuralGCM. *Geophysical Research Letters*, in revision.
2. **Ding, X.**, Tang, Q., Chen, G., Ke, Z., Zhang, J., & Shi, Y. (2026). Mechanisms for stratospheric aerosol injection by wildfires: Roles of circulation. *Journal of Geophysical Research: Atmospheres*, under review.
1. Wang, L., Chen, G., **Ding, X.**, Yang, H., & Domeisen, D. I. V. (2026). Low-frequency stratospheric variability and its link to surface climate: Insights from reanalysis and multi-century coupled climate model simulations. *Weather and Climate Dynamics*, in revision.

## HONORS AND AWARDS

**Jacob A. Bjerknes Award**, Department of AOS, UCLA — Jan 2025
*Dissertation award for outstanding research achievements by graduate students.*

**Travel Support**, DynVar/SNAP Workshop — Jul 2023

**Dissertation Year Fellowship**, UCLA — Jun 2023

**Paul M. Furukawa Memorial Fellowship**, Department of AOS, UCLA — Sep 2022

**NCAR ASP Colloquium Travel Award** — Jun 2022

**Outstanding Bachelor's Thesis**, Lanzhou University — May 2019

**Outstanding Student Scholarship**, Lanzhou University — Jun 2017

## SEMINARS

Climate Program CWSS Seminar, Lawrence Livermore National Laboratory — Aug 2024

AOS Department Seminar, University of California, Los Angeles — Apr 2024

UNIL Atmospheric Processes Group Seminar, University of Lausanne, Switzerland — Feb 2024

## SELECTED CONFERENCE PRESENTATIONS

**2026.** Stratospheric Wave Predictability Enhances Surface Forecasts over North America. AGU Annual Meeting. *Oral.*

**2026.** Correcting the Stratosphere Improves S2S Forecasts in NeuralGCM. AGU Annual Meeting. *Poster.*

**2025.** Stratospheric Predictability and Its Surface Linkages in S2S Forecasting. AGU Fall Meeting. *Oral.*

**2025.** Demystifying the Drivers of the Spring Warming Asymmetry between Eurasia and North America. AGU Fall Meeting. *Oral.*

**2024.** The Masked Effect of Anthropogenic Aerosols in Driving Zonally Asymmetric Warming Trends in Boreal Spring. AGU Fall Meeting. *Poster.*

**2024.** Strong Stratospheric Wave Events Precede North American Cold Extremes. AMS Annual Meeting. ***Invited oral.***

**2023.** The Impact of QBO on the Surface Signature of Strong Stratospheric Wave Events. AGU Fall Meeting. *Poster.*

**2023.** Stratospheric Wave Precursor of Cold Events over North America. DynVar/SNAP Workshop. *Oral.*

**2023.** Causality between Extreme Stratospheric Wave Activity and Cold Extremes over North America. S2S Summit. *Poster.*

**2023.** Assessing Stratosphere-troposphere Coupling of Extreme Stratospheric Wave Activity in CMIP6 Models. EGU General Assembly. *Oral.*

**2022.** Examining the Causality between Stratospheric Planetary Wave Activity and North American Cold Extremes. AGU Fall Meeting. *Oral.*

**2022.** NCAR ASP Colloquium Workshop. *Poster.*

**2022.** Distinct North American Cooling Signatures Following the Zonally Symmetric and Asymmetric Modes of Stratospheric Variability in the Northern Hemisphere Winter. AMS Annual Meeting. *Oral.*

## TEACHING EXPERIENCE

**AOS 3 — Meteorology and Extreme Weather** Spring 2023
Teaching Assistant, University of California, Los Angeles

**AOS C170/C227 — Advanced Dynamics and Synoptic Meteorology** Winter 2023
Special Reader, University of California, Los Angeles

**AOS 51 — Fundamentals of Climate Science** Spring 2022
Teaching Assistant, University of California, Los Angeles

## PROFESSIONAL SERVICE

### Journal Peer Review

*Nature Geoscience; Nature Communications; npj Climate and Atmospheric Science; Geophysical Research Letters; Journal of Geophysical Research: Atmospheres; Journal of Climate; Atmospheric Chemistry and Physics; Climate Dynamics; Weather and Climate Dynamics; Journal of Meteorological Research.*

### Conference and Departmental Service

Organizing committee, 8$^{th}$ California Geophysical Fluid Dynamics (CalGFD) Meeting 2026

Outstanding Student Presentation Awards reviewer, AGU Fall Meeting 2024

Poster contest judge, AMS Student Conference 2024

Session chair, DynVar/SNAP Workshop, Munich, Germany 2023

Volunteer, S2S Summit, University of Reading 2023

Student representative, AOS 270 seminar committee, UCLA 2022–2023

Volunteer, Exploring Your Universe, UCLA 2021, 2022

## MENTORING

Graduate mentees: Anna Coomans (2026–), Rui Zhang (2024), Jacob Chen (2022–2023).

Undergraduate mentee: Claire Fu (2021). All mentees co-supervised with Prof. Gang Chen.